UK E-commerce Sales Performance Analysis 2018–2019

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> **Important**
>
> Please run `renv::restore()` in the R console before attempting to render this document.

Executive Summary

This report analyses the retail transaction data of a UK based e-retailer selling gifts and house ware for all ages, to identify sales and revenue patterns across the United Kingdom and international markets during the Jan 2019 - Dec 2019 period. The data set was cleaned by removing duplicate records and creating new variables to support analysis. The primary focus of this report was to compare transaction volume and the revenue generated within the UK and in the international market, with a secondary focus on identifying countries outside the UK that may represent viable growth opportunities. Results indicated that while the majority of transaction volume and revenue was generated by the UK, a few countries, such as the Netherlands and Ireland (EIRE), had a disproportionately higher revenue against transaction volume, which may merit further investigation.

Introduction

Research Design and Purpose

> **Purpose**: examine the sales performance of a UK-based online retail business over a 13-month period from December 2018 to December 2019.

Specifically, the analysis investigates transaction volume and revenue trends over time, and evaluates whether performance differs between the United Kingdom and international markets.

This study first adopts a descriptive analytical approach, utilising transactional records to summarise and visualise patterns in the data. Furthermore, further business insights are further investigated to identify key values that would inform future operational and marketing decisions.

Methodology

Data Collection and Sources

This report utilizes secondary data sourced from *Kaggle*, a prominent open-source data repository (Ramos (n.d.)). The dataset comprises one year of historical sales transactions from Dec 2018 to Dec 2019 from a UK-based online retailer. This retailer is headquartered in London, specializing in gifts and homewares for both adults and children.

The business operates on a global scale and serves a dual customer base. Operations include **both direct-to-consumer (B2C) sales**, where individuals purchase for personal use, **and business-to-business (B2B) transactions**, where small businesses purchase items in bulk for retail distribution. Consequently, the transaction data is expected to exhibit diverse purchasing patterns, reflecting both individual consumer behavior and wholesale buying cycles.

Structurally, the dataset contains **536,350 observations** and **8 variables**. Each record represents the quantity of a single product sold as part of a transaction; a transaction can have multiple products.

Table 1: Raw Dataset Variables Descriptions

Variable	Type	Description
`TransactionNo`	Character	Unique transaction identifier (prefix 'C' = cancellation)
`Date`	Character	Date of transaction
`ProductNo`	Character	Unique product identifier
`ProductName`	Character	Name of product sold
`Price`	Numeric	Unit price in GBP (£)
`Quantity`	Numeric	Units sold (negative values indicate returns)
`CustomerNo`	Numeric	Unique customer identifier
`Country`	Character	Country of the customer

Preparing Dataset for Analysis

To ensure data integrity and facilitate robust analysis, the raw dataset underwent a systematic cleaning and transformation process. Initially, variable names were standardized for consistency, and exact duplicate records were identified and removed to prevent observation bias and the double-counting of revenue.

- **Standardization & Deduplication**: Column headers were standardized, and duplicate records were removed from the dataset.
- **Financial Feature Engineering**: A discrete `revenue` variable was calculated for each line item (Price $\times$ Quantity).
- **Temporal Decomposition**: The original date string was parsed into a datetime format and decomposed into `year`, `month`, and `year_month` fields to facilitate seasonal and time-series analysis.
- **Transaction Classification**: A `transaction_type` indicator was created to isolate legitimate sales from returns. Any record with a negative quantity or a transaction ID starting with **"C"** was flagged as a `Cancelled/Return`.

Following the deduplication and feature engineering phases, the final prepared dataset consists of 531,150 observations across 13 variables.

Analysing Sales Trends over Time

In order to identify trends in both number of transactions and total revenue for the UK and non-UK markets, all transactions in the dataset were first assigned one of these two regions. This allowed for the creation of Figure 1, a collection of time-series plots for transactions and revenue divided by region.

These plots allow for the easy visual identification of market trends across the 13-month period we have data for.

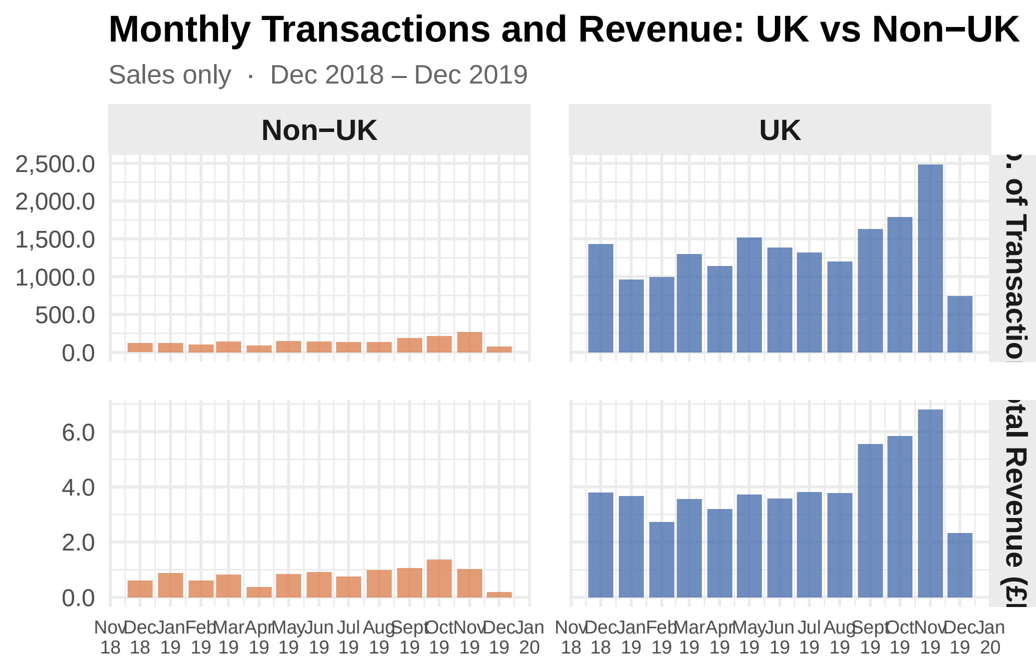

Figure 1: Monthly Transaction and Revenue in UK vs non-UK countries in 2018-2019 period

Results

Number of orders and total Volume by month

As presented in Figure 1, The UK market consistently dominated in both number of transactions and revenue throughout the recorded period. A pronounced upward trend is observed from September 2019 onwards, with transactions and revenue in the UK peaking in November 2019 at nearly 2500 orders and approximately £6.8M respectively, likely driven by seasonal demand ahead of the holiday period.

The Non-UK market, while significantly smaller in absolute scale, demonstrated a gradual growth trajectory across 2019 with the upward trend merely mirror UK market. This suggests a slow but steady expansion of the international customer base.

Figure 2 shows the top non-UK countries by revenue amount, and also hints at an interesting idea; that the retailer is performing very well in some countries despite relatively low numbers of transactions. This will be covered in more depth in Section .

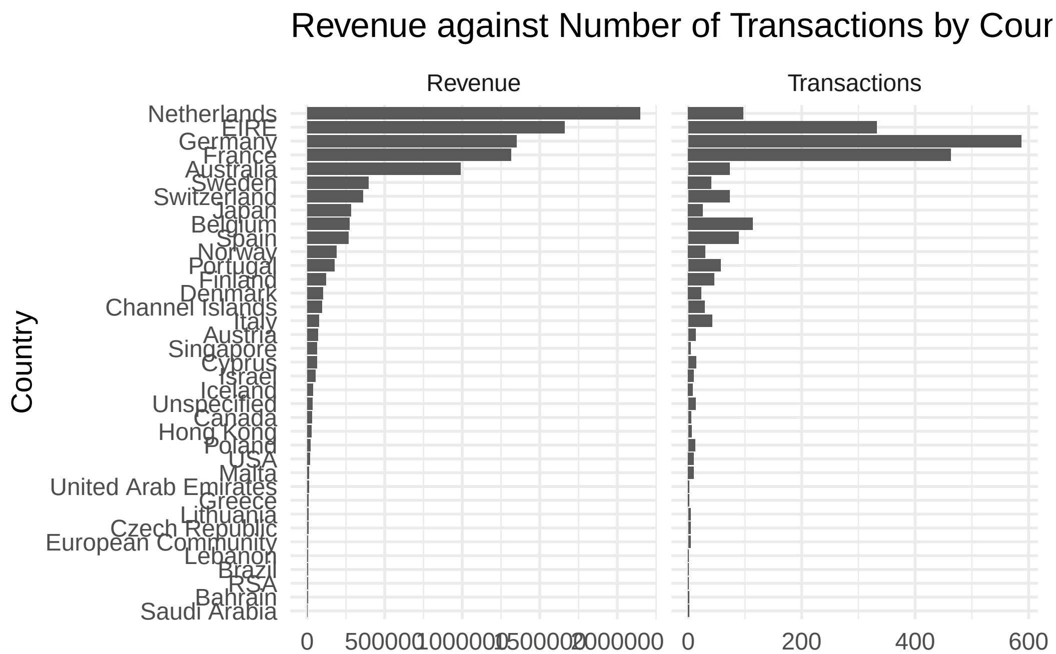

Figure 2: Transactions and Revenue for Non-UK Countries

General Quantitative metrics

As seen in Figure 1, both the UK and Non-UK markets reached their peak transaction volumes and revenue towards the end of 2019. Referencing Table 2, the UK reached 2487 transactions and £6.81M in revenue in November 2019, while Non-UK countries peaked at 266 transactions in the same month. This Q4 surge is consistent with seasonal holidays like Christmas and New Year.

Notably, Non-UK maximum revenue occurred in October 2019 (£1.37M) rather than November despite November having more transactions, suggesting a higher average basket value per order in October.

The minimum values for both regions fall in December 2019 across all metrics. This might not be a genuine business downturn but rather a data artefact, as December 2019 represents an incomplete month in the dataset.

Table 2: Monthly Transaction and Revenue in UK vs non-UK countries in 2018-2019 period

	Region	
Metric	**Non-UK**	**UK**
Total Transactions	1,882	17,908
Mean Transactions/Month	145	1,378
Min Transactions/Month	73	744
Max Transactions/Month	266	2,487
Total Revenue (£)	10,434,508.94	52,346,877.60
Mean Revenue/Month (£)	802,654.53	4,026,682.89
Min Revenue/Month (£)	182,853.09	2,329,216.43
Max Revenue/Month (£)	1,374,514.18	6,806,782.19

Countries' basket value

Figure 3 presents basket size statistics across top 10 countries with the biggest basket and the UK. Netherlands, Australia and Singapore recorded the highest mean basket values at £22888, £16055 and £15870 respectively. Notably, these countries achieved these values from a very small number of transactions, especially Singapore had only 4 transactions, this pattern could indicate wholesale or B2B purchasing behaviour rather than individual retail orders.

Cancelled orders insight

Figure 4 shows that the USA has the highest cancellation rate at 29.63%, followed by Malta (9.4%) and Japan (9.16%), all well above the UK benchmark of 1.52%. However, despite its low rate, the UK incurs the largest revenue loss at £2509.3K due to its transaction number. Among Non-UK markets, EIRE (Ireland) records the highest revenue lost at £52.8K, making it the most financially impactful cancellation market outside the UK.

Dicussion

The results indicates that this business is operationally strong in its domestic market but has significant untapped potential internationally.

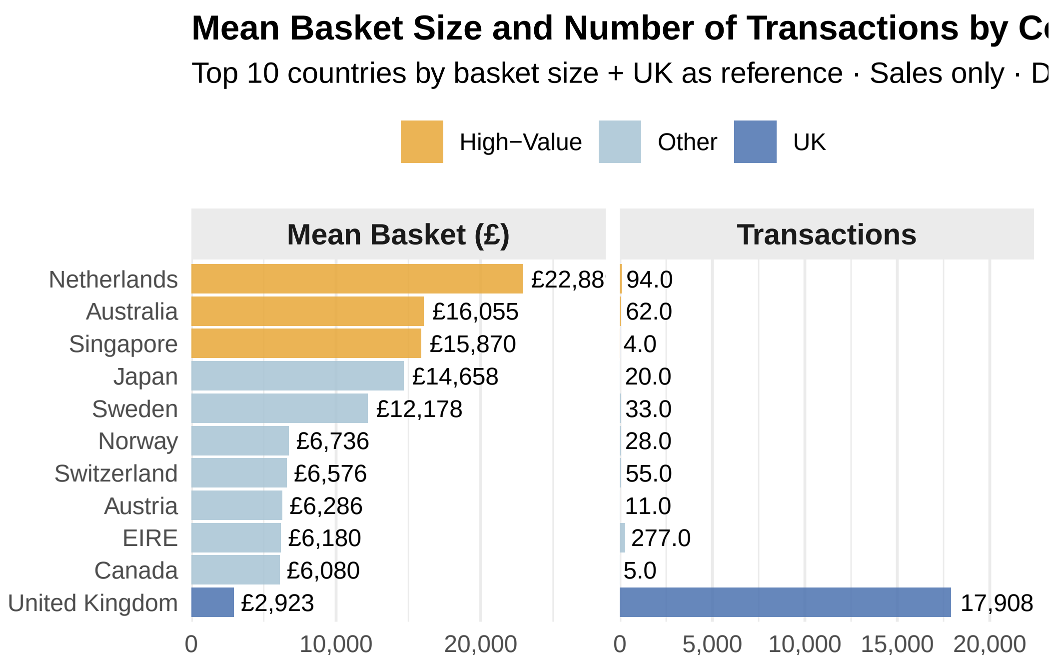

Figure 3: Metrics for Basket Size per transaction by country in 2018-2019 period

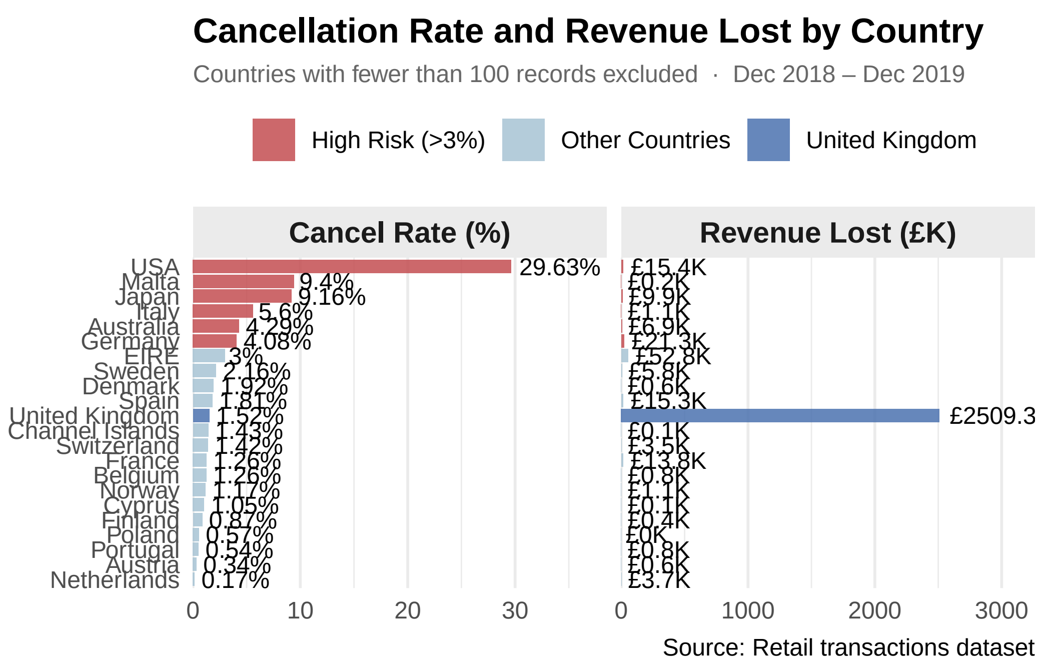

Figure 4: Cancellation rate and Revenue lost by each Country

General market is Seasonal

The dominant Q4 seasonality observed in Figure 1 suggests the business is **heavily reliant on the festive season period** to drive annual performance. While this concentration generates strong peak revenue, it also **exposes the business to seasonal risk**.

High value international B2B market

The basket size analysis in Figure 3 reveals a more complex international picture. Despite lower transaction volumes, most **Non-UK customers spend nearly exponentially more per order comparing to the UK**. Countries such as the Netherlands, Singapore, and Australia demonstrate average basket sizes of approximately £16000, mostly indicate that they belong to the **B2B customer segment**. Rather than casual retail consumers, these clients are bulk purchasers placing infrequent, high-value orders, making them disproportionately valuable relative to their transaction volume.

Barrier regarding international sales

However, Figure 4 highlights a key operational challenge in international markets. The cancellation rate at the USA (29.6%), Japan (9.2%) and Germany (4.1%) all exceed the UK benchmark of 1.52%, suggesting that international fulfilment reliability remains inconsistent. While EIRE represents the largest Non-UK revenue loss at £52.8K, the high-rate markets like the USA likely reflect **demand uncertainty** or **logistics hindrance** rather than product dissatisfaction.

Recommendation

Reduce seasonal dependency: The business should invest in mid-year promotional campaigns, particularly in months of February and April while they showed the modest number, to reduce the financial risk of a single underperforming holiday season.

Formalise and grow the international wholesale channel: The high basket sizes recorded in the Netherlands, Australia and Singapore signal an existing wholesale customer base. Rather than treating these as ordinary transactions, the business should *develop a dedicated B2B programme with volume-based pricing, account management and tailored outreach.*

Address international fulfilment gaps before scaling: The elevated cancellation rates in some countries suggest that the operational infrastructure for international orders is not yet mature. Before investing further in these markets, the business should investigate the *root causes of cancellations: whether driven by shipping delays, product mismatches, or payment friction.* The Netherlands, with a near-zero cancellation rate of 0.17%, offers a useful operational benchmark for what reliable international fulfilment looks like.

References

Firke, S. (2024). *Janitor: Simple tools for examining and cleaning dirty data*. https://doi.org/10.32614/CRAN.package.janitor

Grolemund, G., & Wickham, H. (2011). Dates and times made easy with lubridate. *Journal of Statistical Software*, *40*(3), 1–25. https://www.jstatsoft.org/v40/i03/

Ramos, G. (n.d.). *E-commerce Business Transaction*. Retrieved June 6, 2026, from https://www.kaggle.com/datasets/gabrielramos87/an-online-shop-business

Ushey, K., & Wickham, H. (2026). *Renv: Project environments*. https://doi.org/10.32614/CRAN.package.renv

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., … Yutani, H. (2019). Welcome to the tidyverse. *Journal of Open Source Software*, *4*(43), 1686. https://doi.org/10.21105/joss.01686

Wickham, H., Pedersen, T. L., & Seidel, D. (2025). *Scales: Scale functions for visualization*. https://doi.org/10.32614/CRAN.package.scales

Xie, Y. (2025). *Knitr: A general-purpose package for dynamic report generation in R*. https://yihui.org/knitr/

Zhu, H. (2024). *kableExtra: Construct complex table with 'kable' and pipe syntax*. https://doi.org/10.32614/CRAN.package.kableExtra